

Virtual Disk Image Analysis & Troubleshooting

When testing the Aegis-OS system images using QEMU, it is occasionally necessary to inspect the raw virtual disk to verify structural integrity, particularly if the bootloader fails to execute properly. We utilize **TestDisk** to analyze the partition structure and verify boot sector signatures.

Diagnosing Boot Sector Errors

If the system image fails to boot or is unrecognized, follow these steps to analyze the disk structure:

1. Select Partition Table Type

Launch TestDisk and select the virtual disk (`/dev/sda` in this environment). Because the image is being built from scratch and may not yet have a standard partition table written to it, select **[None] Non partitioned media**.

```
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/sda - 407 MB / 388 MiB (R0) - Virtual Disk

Please select the partition table type, press Enter when done.
[Intel  ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax  ] Humax partition table
[Mac    ] Apple partition map (legacy)
>[None  ] Non partitioned media
[Sun    ] Sun Solaris partition
[XBox   ] Xbox partition
[Return] Return to disk selection

Hint: None partition table type has been detected.
```

Selecting the appropriate media type for the raw virtual disk.

2. Initiate Disk Analysis

From the disk menu, select **[Analyse]** to scan the current partition structure and search for any lost or misaligned partitions. Correct disk geometry is critical here, but TestDisk will handle standard 512-byte sector sizes for the virtual image.



Initiating the analysis process on the 407 MB virtual disk.

3. Missing Boot Signature (0xAA55)

During the creation of the bootloader, ensuring standard compatibility (like the Multiboot 1 header) is crucial. A common error during early development is the absence of the Master Boot Record (MBR) boot signature. If TestDisk returns the error `Partition sector doesn't have the endmark 0xAA55`, the binary is missing the standard 55 AA magic number at the end of the boot sector (byte 510 and 511).

```
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Disk /dev/sda - 407 MB / 388 MiB - CHS 49 255 63 (R0)
Current partition structure:
  Partition      Start      End    Size in sectors

Partition sector doesn't have the endmark 0xAA55

**Primary bootable  P=Primary  L=Logical  E=Extended  D=Deleted
>[Quick Search]
      Try to locate partition
```

TestDisk successfully identifying the missing 0xAA55 boot signature.

4. Search Results & Resolution

If a Quick Search yields **No partition found or selected for recovery**, it confirms the image is entirely raw or the partition table is entirely unwritten.

```
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Disk /dev/sda - 407 MB / 388 MiB - CHS 49 255 63 (R0)
  Partition      Start      End    Size in sectors

No partition found or selected for recovery

>[ Quit ] [ Return ]
      Return to main menu
```

result indicating no recoverable standard partitions.

Final

Next Steps:

- Verify the linker script to ensure the bootloader header and signature are placed at the correct memory offset.
- Recompile the system image to include the necessary `0xAA55` endmark before the next QEMU test run.